Conditions and Function

Tasks

In this task you are asked to construct a table showing sales information of Ice Cream Sales accross Australia.

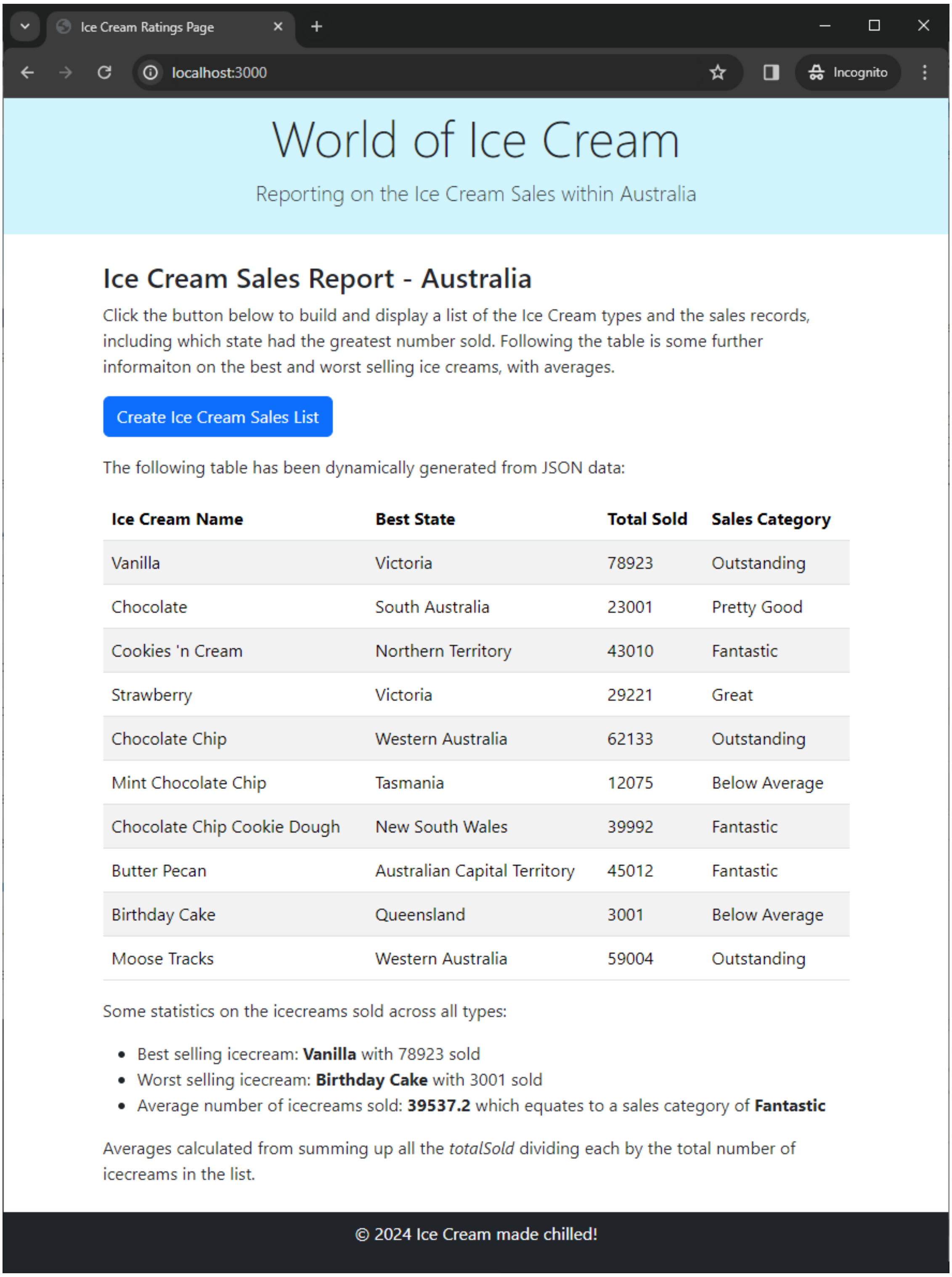
You will be provided with some raw data for the set of ice creams, which includes fields such as the Ice Cream Name, the State the Number Sold and the Average User rating.

In completing this task, the code used to dynamically build the table should be allocated within a javascript function. This function is called when the user clicks a button.

The initial webpage should appear as shown below:

Initial page view

When the user clicks the button to generate the table, it should appear as shown in the following two images below:



Ice Cream Ratings Page

localhost:3000

Incognito

World of Ice Cream

Reporting on the Ice Cream Sales within Australia

Ice Cream Sales Report - Australia

Click the button below to build and display a list of the Ice Cream types and the sales records, including which state had the greatest number sold. Following the table is some further informaiton on the best and worst selling ice creams, with averages.

Create Ice Cream Sales List

The following table has been dynamically generated from JSON data:

Ice Cream Name	Best State	Total Sold	Sales Category
Vanilla	Victoria	78923	Outstanding
Chocolate	South Australia	23001	Pretty Good
Cookies 'n Cream	Northern Territory	43010	Fantastic
Strawberry	Victoria	29221	Great
Chocolate Chip	Western Australia	62133	Outstanding
Mint Chocolate Chip	Tasmania	12075	Below Average
Chocolate Chip Cookie Dough	New South Wales	39992	Fantastic
Butter Pecan	Australian Capital Territory	45012	Fantastic
Birthday Cake	Queensland	3001	Below Average
Moose Tracks	Western Australia	59004	Outstanding

Some statistics on the icecreams sold across all types:

- Best selling icecream: **Vanilla** with 78923 sold
- Worst selling icecream: **Birthday Cake** with 3001 sold
- Average number of icecreams sold: **39537.2** which equates to a sales category of **Fantastic**

Averages calculated from summing up all the *totalSold* dividing each by the total number of icecreams in the list.

© 2024 Ice Cream made chilled!

List of Australian Cities

You are provided with the raw data for the ice cream sales in the form of a JSON object, that contains an array of `icecreams`. This JSON object is shown below:

```
const icecreamSalesListJSON = {
  icecreams: [
    { name: "Vanilla", bestState: "VIC", totalSold: 78923 },
    { name: "Chocolate", bestState: "SA", totalSold: 23001 },
    { name: "Cookies 'n Cream", bestState: "NT", totalSold: 43010 },
    { name: "Strawberry", bestState: "VIC", totalSold: 29221 },
    { name: "Chocolate Chip", bestState: "WA", totalSold: 62133 },
    { name: "Mint Chocolate Chip", bestState: "TAS", totalSold: 12075 },
    { name: "Chocolate Chip Cookie Dough", bestState: "NSW", totalSold: 39992 },
    { name: "Butter Pecan", bestState: "ACT", totalSold: 45012 },
    { name: "Birthday Cake", bestState: "QLD", totalSold: 3001 },
    { name: "Moose Tracks", bestState: "WA", totalSold: 59004 },
  ],
};
```

> This ice cream data has been provided as a Javascript Array of objects which you may use. This is in a file named `icecreamData.zip` that is downloadable from the *Task7.3D* Ontrack page, in the **Task Details -> Resources** link at the bottom right hand side of the web page.

NOTE: The JSON object provided has some fields that **are** shown in the table, while the last column is **different** and not provided in the raw data. In completing this task, you will be required to write a support javascript function that takes an input (parameters) and return the processed result which is used to populate the table.

For example, the **Best State** is provided in the data in an abbreviated form, which needs to be presented in the second column in full-form, i.e., `VIC` should be presented as `Victoria`. A javascript function that takes the *state* as input and returns the full name as a string could be used. An example of such a function outline is shown below:

```
function fullStateName(state) {
  // Should use the input 'state' and return a string
  // with the full name (HINT: a switch() statement)
}
```

In addition the **Sales Category** field is not provided in the data so needs to be constructed and displayed in the final column. A javascript function that takes the *user rating* as input and returns a *sales category* string could be used. An example of such a function outline is shown below:

```
function salesCategory(userRating) {
  // Should use the input 'userRating' and return a string
  // with the full name (HINT: a switch() statement)
}
```

The final column in the table is **Sales Category** which is dependent on the *user* *rating* value. The following table details how the activity should be chosen:

Ice Cream Sales Range	Category Name
0 <= userRating < 15000	‘Below Average’
15000 <= userRating < 25000	‘Pretty Good’
25000 <= userRating < 35000	‘Great’
35000 <= userRating < 55000	‘Fantastic’
userRating >= 55000	‘Outstanding’

In calculating the **Sales Category** field, a javascript function should be developed that takes as input the *user rating* and using a cascading *if / then / else if* condition checks, returns the text (*string*) for the appropriate *category* from the table above (i.e., `function salesCategory(userRating) { .. }`).

Minimum / Maximum / Average Summary

The final section of the dynamically created table are the fields to show the various interesting features from the data. This includes:

- The ice cream type with the greatest number sold and form which state
- The ice cream type with the least number sold and form which state
- The calculated *average number of ice creams sold* and the *average sales rating* that this would equate to. This information should be displayed below the table.

Hints:

- Use the JSON array of objects provided to store the Ice Cream details and information.
- Define a Javascript function (i.e., `salesCategory( userRating )`) to return the *Sales Category* when provided with the ice cream's sale total (*totalSold* value).
- Use a table to organize the displayed data.
- Use additional variables to keep a total of the *icecreams sold*, the *index* of the best & worst selling ice creams; these can be updated in the loop as you build the html to display the ice cream's row.

What will you submit?

You should submit:

- Screen-shot(s) of the web page showing the final table.
- `HTML` base file
- The `JavaScript` code to construct the table.